

SECTION 1: INDENT & CHARTERING PARAMETERS

Commodity Type:	Prime Hard Coking Coal	Parcel Volume:	75,000 MT
Loading Port:	Gladstone	Discharge Port:	Paradip
Vessel Class:	Kamsarmax	Operating Draft:	13.67 m
Distance Nautical Miles:	4850 NM	Port Clearance Status:	APPROVED (Clear)

SECTION 2: AI FREIGHT VALUATION & BENCHMARK CEILINGS

Admiralty Cost Floor	Target Rate (P50)	P90 Ceiling (Max Cap)	Portfolio Savings
\$8.15/MT	\$20.95/MT	\$21.09/MT	INR 13.76 Cr

SECTION 3: SHAP EXPLAINABLE AI DECOMPOSITION (STATUTORY AUDIT PROOF)

Model Baseline Prior Expected Value: \$18.66 / MT | Explainer Algorithm: LightGBM TreeExplainer

Feature / Market Driver	Economic Significance	Marginal Impact (\$/MT)
VLSFO SINGAPORE	Singapore 0.5% VLSFO Bunker Fuel Spot Quote	\$-0.067 / MT
BALTIC P3A	Baltic Exchange P3A (Pacific RV) Capesize/Panamax T/C	\$-0.034 / MT
BUNKER MA7	7-Day Short-Term Exponential Moving Average of Bunker	+\$0.217 / MT
BALTIC MA30	30-Day Long-Term Moving Average Baltic Freight Trend	+\$0.028 / MT
FFA 30D PAPER	Forward Freight Agreement (FFA) 30-Day Paper Derivative Curve	+\$2.140 / MT

SECTION 4: TSD TENDER & CHARTERING DIRECTIVE

Procurement Timing: EXECUTE NORMAL TENDER: Stable market within expected P50 trajectory.

Contract Allocation: Absorb into 6-Month COA Tranche (Saves ~Rs.0.79 Cr on parcel)

Port Demurrage Risk: SimPy Expected Wait: 0.77 days | Est Demurrage Exposure: \$0.00

Chief General Manager (Shipping)
SAIL TSD Kolkata

Vigilance & Compliance Officer
Government Audit Reviewer

AI Decision System Seal
Naavai Team F6 PS26006

STATUTORY PROOF: WHY TARGET RATE \$20.95/MT & P90 CEILING WERE ESTABLISHED		
1	CHECKPOINT 1: Naval Physics Thermodynamic Floor Quantitative Finding: \$8.15/MT Admiralty speed-power formula across 4,850 NM at VLSFO \$620/MT establishes thermodynamic fuel+OPEX breakeven. Shipowners will refuse fixtures below \$8.15/MT.	[PHYSICAL LOWER BOUND]
2	CHECKPOINT 2: Nautical Draft & UKC Clearance Quantitative Finding: 13.67m Draft OK - Vessel & parcel cleared for docking Net Under-Keel Clearance guarantees safe passage without seabed grounding.	[CLEARANCE APPROVED]
3	CHECKPOINT 3: AI Machine Learning Quantile Pricing Quantitative Finding: \$20.95/MT LightGBM Quantile regression models forward freight trajectory from Baltic P3A (\$19,500.0/day), FFA paper (\$21.00/MT), and SHAP drivers. Target P50 rate of \$20.95/MT reflects fair market val	[MARKET EQUILIBRIUM]
4	CHECKPOINT 4: Port Congestion & Demurrage Simulation Quantitative Finding: 0.77d Wait SimPy stochastic discrete-event simulation at Paradip projects 0.77 days anchorage turnaround. Demurrage rate of \$21,000/day is factored into landed cost.	[CONGESTION BUFFERED]
5	CHECKPOINT 5: MILP Strategic Procurement Allocation Quantitative Finding: COA Tranche EXECUTE NORMAL TENDER: Stable market within expected P50 trajectory. OR-Tools SCIP solver confirms absorbing into COA tranche yields 70.79 Cr savings on this cargo while protecting SAIL inve	[OPTIMAL DIRECTIVE]

Chief General Manager (Shipping) SAIL TSD Kolkata	Vigilance & Compliance Officer Government Audit Reviewer	AI Decision System Seal Naavai Team F6 PS26006
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